

EQUITY ARENA

High-Frequency Real-Time Tournament Trading Simulation Platform

Equity Arena is an enterprise-grade stock exchange competition platform built for collegiate and corporate trading tournaments. Engineered with a real-time Geometric Brownian Motion (GBM) microsecond matching engine, server-authoritative WebSocket streams, atomic dual-ledger transactions, administrative market intervention controls, and institutional interactive playbooks.

Platform Version	v8.0.4 Pre-Event Verified (Clean Git Main)
Author & Lead Architect	Avadhoot Savle
Admin Access Point	avadhoot@krishna.kavya (Hardened Dual-Factor / Fully Masked)
Core Currency Unit	Ignite Credits (IC) • Starting Balance: 20,000.00 IC
Matching Engine	Dual-Layer Continuous GBM + 15M Macro Volatility Swings
Tech Stack	React 18, Vite, Node.js, Express, Socket.IO, Prisma ORM, SQLite
Audit Status	100% Verified (25/25 Integration Tests Passing)

EXECUTIVE SYSTEM ARCHITECTURE

Equity Arena decouples market generation from client interaction. A server-authoritative engine ticks independently every 6 seconds, producing realistic order-book depth, sector correlations, macro cycle drift, and sentiment-triggered news shocks. Traders experience zero-latency spot execution, automated limit orders, portfolio equity recalculation, and a live broadcast wire.

1. AUTHENTICATION & ACCESS CONTROL

The security subsystem governs participant access and admin authorization through cryptographically signed JWT tokens with role-based route guards.

Trader Frictionless Sign-In	Traders log in using their registered Email and Phone Number (which doubles as their initial cryptographic password). No cumbersome password setup or email verification links to delay tournament start times.
Hardened Admin Cloaking	Admin controls are completely hidden from normal users. Unlocked only via a secret 5-click easter egg gesture on the header emblem. The admin email field is fully masked ('••••••••') with autocomplete disabled to prevent unauthorized peeking.
Sliding Window Rate Limiter	In-memory sliding window rate limiting prevents rapid click spamming and brute-force attacks (400ms cooldown on trade executions, 5 failed attempts per 15 mins lock on admin credentials).
Dual-Layer JWT Authorization	Dual-scope JWT authentication distinguishing TRADER (24-hour expiration) from ADMIN (2-hour high-security expiration). Invalid or expired tokens return clean 401 unauthenticated signals to auto-purge localStorage.
Full Roster Import & Management	Admins can manually register participants or bulk-import hundreds of traders via XLSX/CSV spreadsheets with fuzzy column mapping (recognizes 'Student Name', 'Mobile', 'Email ID', etc.).
One-Click Account Actions	Individual trader top-up (inject IC), password reset to phone number, portfolio reset back to 20,000 IC, and emergency single-click 'Reset All' or 'Delete All' actions for pre-event roster wiping.

2. DUAL-LAYER QUANT MARKET ENGINE

A proprietary quantitative simulation engine that prevents static artificial charts and emulates real institutional equity trading.

Layer 1: Geometric Brownian Motion	Continuous micro-volatility calculated every 6 seconds using standard GBM: $dS = S * (\mu dt + \sigma dW)$. Generates lifelike market noise without unnatural straight lines.
Sector Correlation Matrix	Correlated industry shifts. If an Agriculture or Banking news shock occurs, all related tickers in that sector move in tandem using combined Gaussian noise weights.
Layer 2: 15-Minute Macro Cycles	Scheduled independent multi-tier volatility swings (Low: 30-100 IC, Mid: 100-500 IC, High: 1,000-4,000 IC) that break traders out of equilibrium and test trend-following discipline.
Mean-Reversion Pull	Elastic boundary algorithms ensure that stock prices never collapse to zero or skyrocket infinitely. Outlier stocks face an automatic mean-reverting drift pull back toward baseline.
Granular OHLC Candlestick Feeds	Automated server-side tick bucketing into 5M, 15M, 30M, and 1H candles, recording exact OPEN, HIGH, LOW, and CLOSE metrics without data loss.

Batch Broadcast Efficiency	All 15 listed stocks are updated in a single atomic database transaction and broadcast in a single 'stocks:batch-update' WebSocket packet to maintain sub-10ms network performance.
-----------------------------------	---

3. TRADING EXECUTION & ORDER LIFECYCLE

Real-time spot transactions and automated resting limit orders with zero-brokerage direct fills.

Instant Market Buy (Long)	Immediate spot acquisition of shares at live server price. Atomically debits available IC cash, updates weighted average buy price (avgBuyPrice), and creates verified ledger transactions.
Instant Market Sell (Liquidate)	Immediate cash liquidation of holdings. Verified against available shares (held minus shares escrowed in limit sell orders). Credits cash and logs realized Profit/Loss.
Automated Limit Buy Orders	Resting dip-sniping orders. Escrows (targetPrice * qty) IC in locked reserves. Evaluated on every price tick; triggers instantaneously when market price <= target price.
Automated Limit Sell Orders	Resting profit-target orders. Locks specified share count in escrow. Automatically executes when market price rises to >= target price, releasing IC back to wallet.
Escrow Guard & Double-Spend Defense	Traders cannot double-spend cash or oversell holdings while limit orders are resting. Available balance = walletBalance - lockedFunds. Cancelling an order immediately frees escrowed capital.
In-Flight Order Modification	Traders can view, modify target price/quantity, or cancel resting orders directly from the dedicated 'Limit Orders' tab without page refreshes.

4. TOURNAMENT SESSION & LIFECYCLE CONTROL

Complete event orchestration engine allowing tournament administrators to control match duration, pauses, breaks, and auto-settlement.

Configurable Tournament Clock	Server-authoritative countdown (defaults to 180 minutes / 3 hours) with synchronized client clocks. Backgrounded tabs or network re-connections instantly re-sync to server timestamps.
Intermission & Refreshment Break	Admin can pause the tournament for a custom break (e.g. 10–15 mins) with an announcement broadcast. Freezes market drift, locks order placement, and displays a luxury countdown intermission screen.
Emergency Break Stop & Resume	Admins have instant 'STOP TIMER & RESUME GAME' buttons in both header and banner to cancel a break early and resume live trading instantaneously across all screens.
The 5-Minute Auto-Liquidation Sweep	At T-minus 5 minutes (configurable), trading locks permanently. The engine cancels all pending limit orders and auto-sells every share held by every trader at spot price into pure realized cash.
Closing Bell Lock & Final Rankings	When the master clock hits 00:00:00, all activity ceases. The session transitions to ENDED, locks leaderboards, and triggers an automated database backup.

5. BREAKING NEWS & SENTIMENT STEERING

Catalytic market intelligence system that tests traders' reaction speed, comprehension, and execution discipline.

34 Analyst Pre-Seeded Templates	Rich database of real-world corporate news templates across IT, Banking, Pharma, Green Energy, Defense, Infrastructure, and Commodities with pre-calibrated price drift impacts.
Custom Breaking News Dispatch	Admins can type and broadcast custom breaking bulletins on the fly to react to live event circumstances or test participant adaptability.
Delayed Macro Steering Engine	Triggering a news bulletin broadcasts the headline immediately to all screens while queuing a delayed macro shock (default 30 seconds) that shifts targeted stocks by $\pm 10\%$ to $\pm 35\%$.
Audible Tournament Chimes	Non-intrusive high-definition sound chimes play upon news broadcasts, ensuring participants on any tab immediately look up at the tape.
Persistent News Wire Archive	Traders can access all historical news bulletins in a dedicated sidebar tab to review which sectors are currently in play.

6. ADMIN COMMAND CENTER & INTERVENTION

A dedicated administrative dashboard giving event organizers real-time god-mode control over the entire market.

Percentage-Only Price Interventions	Admins can explicitly surge or crash individual stocks using percentage modifiers (+5%, +15%, -10%, etc.) to stimulate tournament dynamics without causing math artifacts.
Market-Wide Price Shifts	Single-click market crash or bull run tool (/admin/market/adjust-all) allowing organizers to simulate a global recession or liquidity boom across all stocks at once.
Real-Time Trader Drill-Down Modal	Inspect any individual trader's real-time portfolio, cash balance, held positions, unrealized P/L, transaction history, and resting limit orders in a single modal.
Stock Ownership Distribution Map	View a live breakdown of who owns which stocks (/admin/stock-holdings) to see which companies are monopolized or heavily shorted.
Live Trade Tape Feed	Real-time stream of every single buy, sell, limit order, and liquidation executed across the tournament floor as it happens.

7. TRADER TERMINAL & BATTLE PLAYBOOK

The participant terminal is optimized for high-intensity competition, mobile responsiveness, and intuitive trading.

Trading Floor Grid (15 Stocks)	High-contrast card grid showing live spot prices, percentage drift pills, 15-minute mini sparklines, and session high/low bounds with green/red directional flash states.
Interactive Stock Detail Modal	Multi-timeframe price chart (5M, 15M, 30M, 1H) rendered via lightweight Canvas/SVG, trade ticket stepper, quick preset buttons (1, 5, 10, 50, 100), and cost estimator.
Real-Time Net Worth Header	Live header display showing cash balance, locked funds in limit orders, total holdings value, and dynamic count-up animations on wallet credit.
The 'PLAYBOOK' Combat Manual	A built-in tactical field manual with zero AI fluff. Features interactive capital allocation simulators, live trading ticket demos, automated sniper sliders, and news wire demonstrators.
Luxury Intermission Display	During tournament breaks, traders see a full-screen intermission screen with animated gold digital countdown timers, break notes, and safety confirmation that assets are frozen.

8. PUBLIC STADIUM DISPLAY & EVENT AUDIT

Public projector views, automated backups, and complete event audit trails.

Public Projector Leaderboard (/public)	Dedicated full-screen scoreboard designed for big-screen projectors in the arena hall. Displays live rankings, net worth, P/L percentages, and animated podium positions for top 3 traders.
Automatic 15-Minute Database Backups	Background cron daemon that automatically snapshots the entire SQLite database state every 15 minutes to prevent data loss in case of hardware or power failure.
Permanent Ledger Auditability	Every transaction, order fill, cancellation, and admin intervention is logged with millisecond timestamps in the Prisma database for post-event dispute resolution.
Light & Dark Theme Engine	Built-in CSS token design system allowing organizers and participants to toggle between institutional Dark Mode and crisp Light Mode instantly.

FINAL VERIFICATION STATUS	READY FOR TOURNAMENT DEPLOYMENT
Automated Test Suite	25 / 25 Modules Verified (Auth, Orders, Session, GBM, News, Sockets)
Production Build	Vite v6.4.3 • 2047 Modules Compiled Cleanly • 0 Warnings / 0 Errors
Git Synchronized	Pushed to origin/main (Commit 828eec0)